

Checking the general simple wave

Substitution into the one-dimensional shallow water equations

A subscript on a field is a partial derivative. The depth is h , the depth-averaged velocity is u , and g is gravity. Write $c = \sqrt{gh}$, so $h = c^2/g$. The bed is flat, so $z = 0$. This note takes the general simple wave from the companion note and substitutes it into the shallow water equations.

1. The equations

On a flat frictionless bed,

$$\begin{aligned} h_t + (hu)_x &= 0, \\ u_t + uu_x + gh_x &= 0. \end{aligned} \tag{1}$$

In terms of c , using $gh_x = 2cc_x$, these are

$$\begin{aligned} 2c_t + 2uc_x + cu_x &= 0, \\ u_t + uu_x + 2cc_x &= 0, \end{aligned} \tag{2}$$

wherever $c > 0$. The Riemann invariants are $u + 2c$ and $u - 2c$.

2. The candidate

A simple wave makes one invariant constant, so u is an affine function of c . The other invariant is constant along straight characteristics, and an arbitrary differentiable profile φ labels those lines. The two families are

$$u + 2c = m, \quad x = (u - c)t + \varphi(c), \tag{3}$$

and

$$u - 2c = k, \quad x = (u + c)t + \varphi(c). \tag{4}$$

The signs are opposite: the constant invariant belongs to one characteristic family, and the profile φ is carried on the other. In the first family $u - c = m - 3c$. In the second, $u + c = k + 3c$. The depth and velocity are known once the implicit equation has been solved for $c(x, t)$.

The solution is classical only while that equation defines a unique differentiable c . Differentiating in c shows that this holds when

$$\varphi'(c) - 3t \neq 0 \tag{5}$$

in the first family, and $\varphi'(c) + 3t \neq 0$ in the second. When the derivative vanishes, neighbouring characteristics meet and the wave breaks.

The figure is the first family with $m = 2$ and $\varphi(c) = 3c$, between constant states $c = 2/3$ on the left and $c = 1$ on the right. Then $u = 2 - 2c$ and $gh = c^2$. Here $\varphi'(c) - 3t = 3 - 3t$, so the profile is classical for $t < 1$ and breaks at $t = 1$. The curves are $t = 0$ and $t = 2/3$: the transition has narrowed and steepened.

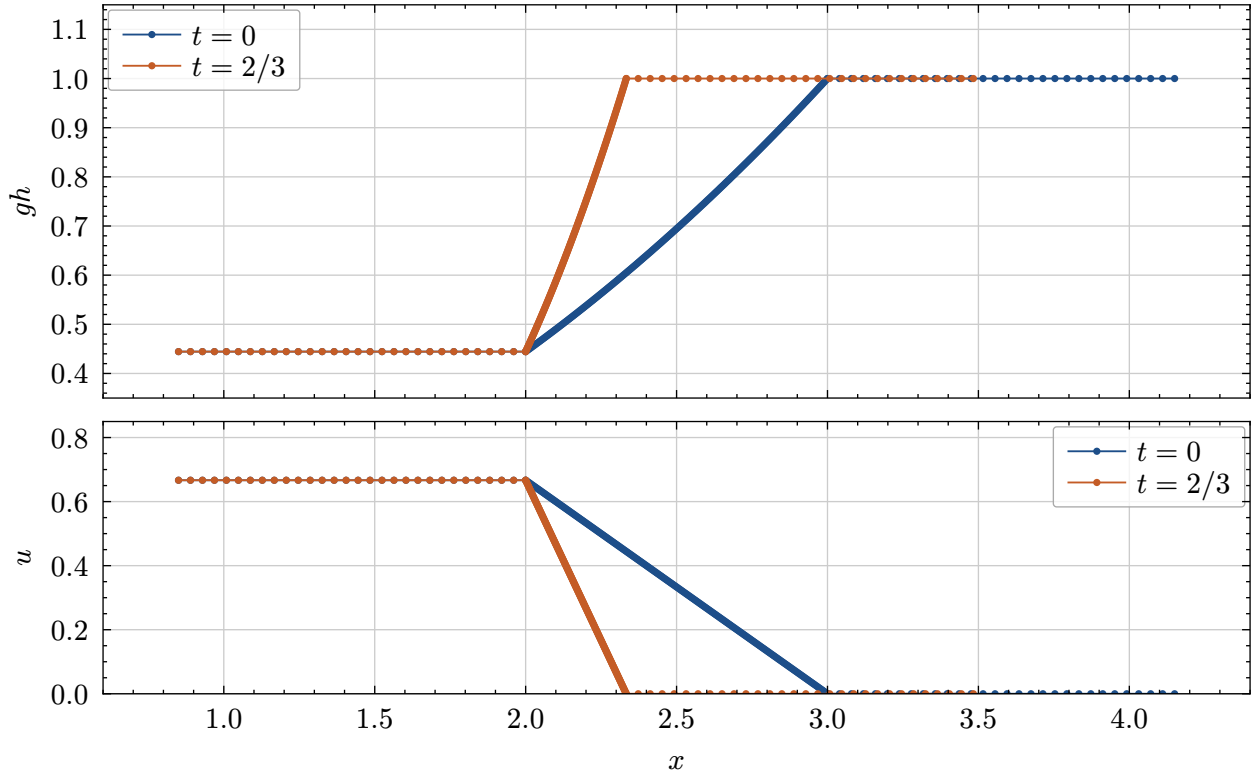


Figure 1: A compressive simple wave of the first family, before breaking at $t = 1$. Depth is drawn as $gh = c^2$.

3. The first family

Fix $u + 2c = m$, so $u = m - 2c$, and let $c(x, t)$ solve

$$x = (m - 3c)t + \varphi(c). \quad (6)$$

Differentiating with respect to x and t , with $\varphi'(c) - 3t \neq 0$, gives

$$c_x = \frac{1}{\varphi'(c) - 3t}, \quad c_t = -(m - 3c)c_x = -(u - c)c_x. \quad (7)$$

Thus $c_t + (u - c)c_x = 0$: the value of c is constant on each characteristic. Since $u = m - 2c$,

$$u_t = -2c_t, \quad u_x = -2c_x. \quad (8)$$

Substitute into the mass equation:

$$2c_t + 2uc_x + cu_x = 2c_t + 2uc_x - 2cc_x = 2(c_t + (u - c)c_x) = 0. \quad (9)$$

Substitute into the momentum equation:

$$u_t + uu_x + 2cc_x = -2c_t - 2uc_x + 2cc_x = -2(c_t + (u - c)c_x) = 0. \quad (10)$$

Both equations hold at every point where $c > 0$ and $\varphi'(c) \neq 3t$.

4. The second family

Fix $u - 2c = k$, so $u = k + 2c$, and let $c(x, t)$ solve $x = (k + 3c)t + \varphi(c)$. The same differentiation, now using $\varphi'(c) + 3t \neq 0$, gives

$$c_t = -(u + c)c_x, \quad u_t = 2c_t, \quad u_x = 2c_x. \quad (11)$$

Mass becomes

$$2c_t + 2uc_x + cu_x = 2c_t + 2uc_x + 2cc_x = 2(c_t + (u + c)c_x) = 0. \quad (12)$$

Momentum becomes

$$u_t + uu_x + 2cc_x = 2c_t + 2uc_x + 2cc_x = 2(c_t + (u + c)c_x) = 0. \quad (13)$$

Both equations hold.

5. Ritter's wave

Ritter's rarefaction is the first family with $m = 2c_L$ and φ constant, centred at the origin, so $\varphi = 0$. Then $x = (2c_L - 3c)t$, hence

$$c = \frac{1}{3} \left(2c_L - \frac{x}{t} \right), \quad u = 2c_L - 2c = \frac{2}{3} \left(c_L + \frac{x}{t} \right), \quad (14)$$

for $-c_L \leq x/t \leq 2c_L$, which is the fan in the dry dam break. A constant φ is the centred case of the same solution.

Where one Riemann invariant is constant, the implicit characteristic formula satisfies the shallow water equations up to breaking.